



Optimization Homework 2

SF1811

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AI level: 1, LLM (grok) was used in this assignment just for study mode and to do certain tedious tasks unrelated to the the actual solution and thinking behind the assignment. We used it for finding what terminology to use for certain variables (eg: A being called the constraint matrix), how to write some symbols in latex, converted the assignment statement to latex, and a tutorial on how to input arguments in to the `linprog` function.

A small manufacturing company produces two products, P_1 and P_2 , in two factories, F_1 and F_2 . Each product can be produced in either factory, but with different costs and resource requirements. The products are sold in three regional markets: M_1 , M_2 , M_3 .

The company must decide:

- how much of each product to produce in each factory, and
- how much to ship from each factory to each market,

in order to maximize total profit, given production capacities, labor constraints, and market demand.

The following data is available.

- **Production capacities**

Factory	Max prod. time (h)	Available labor (h)
F_1	600	500
F_2	800	700

- **Product characteristics**

Product	Time/unit (h)	Labor/unit (h)	Prod. cost (kr)	Selling price (kr)
P_1	2	1	30	60
P_2	3	2	50	90

- **Transportation cost (kr/unit)**

From / To	M_1	M_2	M_3
F_1	4	6	9
F_2	5	4	7

- Upper bound on sales for each market (units)

Market	P_1	P_2
M_1	100	120
M_2	150	100
M_3	200	150

Both production and shipment can be made in arbitrary nonnegative quantities, subject to the conditions above. The company may store products that are not shipped at no cost.

1 Exercise 2.1

Let the following be our decision variables for the LP to maximize profit

$$x_{fp}, \quad f = 1, 2, \quad p = 1, 2,$$

denoting units of product p produced at factory f , and

$$w_{fpm}, \quad f = 1, 2, \quad p = 1, 2, \quad m = 1, 2, 3,$$

denoting units of product p produced at factory f shipped to market m . For the max prod time and available labor constraints we have the following inequalities:

$$2x_{11} + 3x_{12} \leq 600$$

$$x_{11} + 2x_{12} \leq 500$$

$$2x_{21} + 3x_{22} \leq 800$$

$$x_{21} + 2x_{22} \leq 700$$

and for the constraints on units sold we have the following inequalities:

$$w_{111} + w_{211} \leq 100$$

$$w_{112} + w_{212} \leq 150$$

$$w_{113} + w_{213} \leq 200$$

$$w_{121} + w_{221} \leq 120$$

$$w_{122} + w_{222} \leq 100$$

$$w_{123} + w_{223} \leq 150$$

and the objective function is the cost function which is what we're trying to minimize

$$\begin{aligned}
c^T x = & 4(w_{111} + w_{121}) + 6(w_{112} + w_{122}) \\
& + 9(w_{113} + w_{123}) + 5(w_{211} + w_{221}) + 4(w_{212} + w_{222}) + 7(w_{213} + w_{223}) \quad (\text{total transport cost}) \\
& + 30(x_{11} + x_{21}) + 50(x_{12} + x_{22}) \quad (\text{total production cost}) \\
& - 60(w_{111} + w_{112} + w_{113} + w_{211} + w_{212} + w_{213}) \quad (\text{total revenue for P1}) \\
& - 90(w_{121} + w_{122} + w_{123} + w_{221} + w_{222} + w_{223}) \quad (\text{total revenue for P2})
\end{aligned}$$

and we have flow balance constriction where we're allowed to store products but obviously its not possible to ship more than the amount of products we've produced.

$$\begin{aligned}\sum w_{11i} - x_{11} &\leq 0 \\ \sum w_{12i} - x_{12} &\leq 0 \\ \sum w_{21i} - x_{21} &\leq 0 \\ \sum w_{22i} - x_{22} &\leq 0\end{aligned}$$

We'll write our decision variables as a column vector $x \in \mathbb{R}^{16}$ which will have the nonnegativity constraint $0 \leq x$

$$x = \begin{pmatrix} w_{111} \\ w_{112} \\ w_{113} \\ w_{121} \\ w_{122} \\ w_{123} \\ w_{211} \\ w_{212} \\ w_{213} \\ w_{221} \\ w_{222} \\ w_{223} \\ x_{11} \\ x_{12} \\ x_{21} \\ x_{22} \end{pmatrix}$$

We can aggregate all these inequalities in to one $Ax \leq b$ where $A \in \mathbb{R}^{14 \times 16}$ and $b \in \mathbb{R}^{14}$. The bound vector b has its elements in the same order as the RHS of each inequality referenced above. We'll populate the 14×16 constraint matrix A in the code (using symbolic math toolbox we can copy the output of **A** and **b** and write it here)

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & -1 \end{pmatrix}$$

$$b = \begin{pmatrix} 600 \\ 500 \\ 800 \\ 700 \\ 100 \\ 150 \\ 200 \\ 120 \\ 100 \\ 150 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$c^T = (-56 \quad -54 \quad -51 \quad -86 \quad -84 \quad -81 \quad -55 \quad -56 \quad -53 \quad -85 \quad -86 \quad -83 \quad 30 \quad 50 \quad 30 \quad 50)$$

We thereby formulate the LP as

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$$

2 Exercise 2.2

Code solution

Listing 1: main.m

```
1  A = zeros(14,16);
2  %the following 14 lines are the coefficients of each inequality used to populate the
   constraint matrix
3  A(1, 13) = 2; A(1, 14) = 3;
4  A(2, 13) = 1; A(2, 14) = 2;
5  A(3, 15) = 2; A(3, 16) = 3;
6  A(4, 15) = 1; A(4, 16) = 2;
7
8  A(5, 1) = 1; A(5, 7) = 1;
9  A(6, 2) = 1; A(6, 8) = 1;
10 A(7, 3) = 1; A(7, 9) = 1;
11 A(8, 4) = 1; A(8, 10) = 1;
12 A(9, 5) = 1; A(9, 11) = 1;
13 A(10, 6) = 1; A(10, 12) = 1;
14
15 A(11, 13) = -1; A(11, 1) = 1; A(11, 2) = 1; A(11, 3) = 1;
16 A(12, 14) = -1; A(12, 4) = 1; A(12, 5) = 1; A(12, 6) = 1;
17 A(13, 15) = -1; A(13, 7) = 1; A(13, 8) = 1; A(13, 9) = 1;
18 A(14, 16) = -1; A(14, 10) = 1; A(14, 11) = 1; A(14, 12) = 1;
19
20 Aeq = []; % no equality constraints defined
21 beq = [];
22
23 %max prod and available labor constraints
24
25 b = [600;
26      500;
27      800;
28      700;
29      % market cap on units sold constraint
30      100;
31      150;
32      200;
33      120;
34      100;
35      150;
36      % flow balance constraints
37      0;
38      0;
39      0;
40      0];
41
42 c = [4 6 9 4 6 9 5 4 7 5 4 7 30 50 30 50] - [60 60 60 90 90 90 60 60 60 90 90 90 0 0 0
43         0];
44 % lower bound
45 lb = zeros(16,1);
46
```

```
47 x = linprog(c, A, b, Aeq, beq, lb);  
48  
49 max_profit = -c*x;  
50  
51 disp(['max profit is ' num2str(max_profit) 'kr']);
```

3 Exercise 2.3

Results

Running `main.m` which uses the `linprog` function (which uses the dual simplex method) we calculate maximum profit to be 17153.3333 kr

and the optimal decision variables are given by

`x=[100, 0, 0, 120, 13.333, 0, 0, 150, 120, 0, 86.666, 0, 100, 133.333, 270, 86.666]`

So factory 1 and 2 will produce:

F1: 100 P1 products, 133.333 P2 products

F2: 270 P1 products, 87 P2 products

and for allocations between factories and markets we'll have:

F1->M1: 100 P1 products, 120 P2 products

F1->M2: 0 P1 products, 13.333 P2 products

F1->M3: 0 P1 products, 0 P2 products

F2->M1: 0 P1 products, 0 P2 products

F2->M2: 150 P1 products, 86.666 P2 products

F2->M3: 120 P1 products, 0 P2 products

the constraints active in this solution are given by

F1 uses max 600h of production time and 500h of labor time

F2 uses max 800h of production time and 700h of labor time

M1 can sell max 100 P1 and 120 P2

M2 can sell max 150 P1 and 100 P2

M3 can sell max 200 P1 and 150 P2